

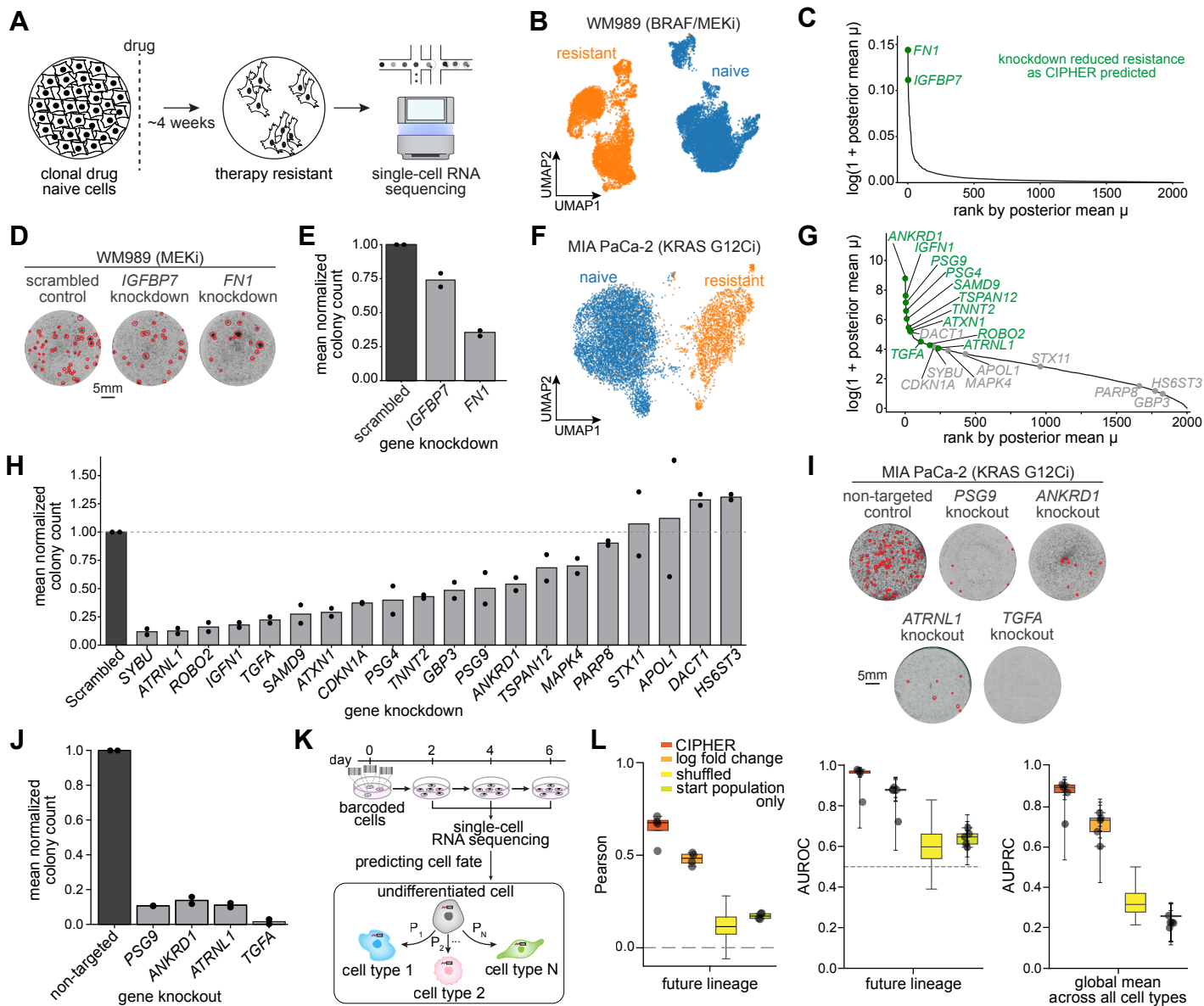


Fig. 7: CIPHER predictions validate experimentally on therapy resistance and cell-fate traits.

a, Schematic of the therapy resistance experiment. Clonal drug-naïve cells were treated with targeted inhibitors to develop resistance; naïve and resistant populations were profiled by single-cell RNA sequencing as control and perturbed conditions. **b**, UMAP of WM989 cells colored by naïve (blue) and BRAF/MEK-inhibitor resistant (orange) populations. **c**, CIPHER ranking of candidate resistance drivers by posterior mean effect size μ . *FN1* and *IGFBP7*, the two top-ranked genes, are indicated. Green indicates genes whose knockdown reduced colony count (d, e) as predicted by CIPHER. **d**, Colony formation for scrambled control, *IGFBP7* knockdown, and *FN1* knockdown in WM989 treated with MEK inhibitor. Scale bar, 5 mm. **e**, Mean normalized colony count for d (n = 2). Scrambled normalized to 1.00. **f**, UMAP of naïve (blue) and resistant (orange) MIA PaCa-2 KRAS-G12C cells after sotorasib treatment. Tested genes labeled. Green indicates genes whose knockdown reduced colony count as CIPHER predicted. **g**, CIPHER ranking of candidate resistance drivers for MIA PaCa-2 by posterior mean μ . Tested genes labeled. Green indicates genes whose knockdown reduced colony count as CIPHER predicted. **h**, Mean normalized colony count after individual knockdown in MIA PaCa-2, ordered by effect (n = 2); scrambled normalized to 1 (dashed line). **i**, Colony formation for non-targeted control and CRISPR-Cas9 knockout of *PSG9*, *ANKRD1*, *ATRNL1*, *TGFA* in MIA PaCa-2. Scale bar, 5 mm. **j**, Mean normalized colony count for I (n = 2). Non-targeted normalized to 1. **k**, Experimental design for lineage prediction in the LARRY hematopoietic differentiation dataset. Barcoded hematopoietic progenitor cells were profiled by single-cell RNA sequencing over multiple time points (days 2, 4, and 6). Early undifferentiated cells are used by CIPHER to predict the probabilities of commitment to each terminal cell fate based on their transcriptional fluctuations. **l**, Performance of lineage prediction methods on the LARRY dataset. Left: Pearson correlation between predicted and observed future lineage commitment probabilities. Middle: area under the receiver operating characteristic curve (AUROC) for identifying the correct future lineage, with the dashed line indicating random performance. Right: area under the precision-recall curve (AUPRC) for the same task. Boxplots compare CIPHER scores, log fold change, shuffled covariance, and a baseline using only the starting population probabilities. Boxes indicate the interquartile range with median, whiskers extend to 1.5X the interquartile range, and individual points denote individual testing folds.